MLOps Pipeline

End-to-End Deployment, Monitoring & Automated Retraining

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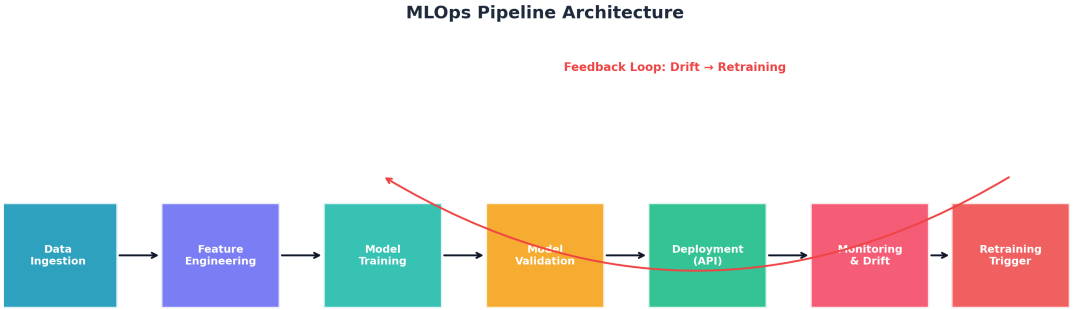
Agenda

1. Pipeline Overview
2. Model Versioning
3. Production Monitoring
4. Drift Detection
5. A/B Testing & Alerts
6. Automated Retraining
7. Conclusions

The MLOps Lifecycle



Continuous Integration / Continuous Deployment (CI/CD)

Project Scope

Demonstrated Capabilities

- Multi-version model registry (3 versions)
- 90-day production simulation (18,000 predictions)
- Real-time accuracy & latency monitoring
- Drift detection (PSI, KS test, calibration)
- Champion vs Challenger A/B testing
- Automated retraining with cost-benefit analysis

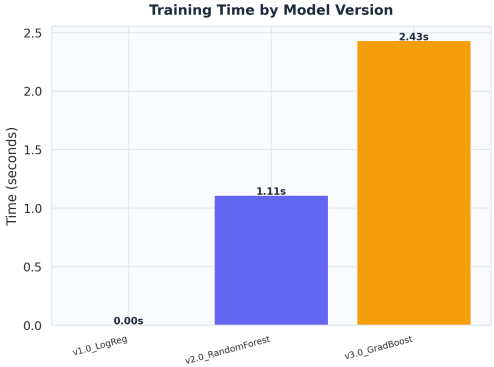
Drift Scenario

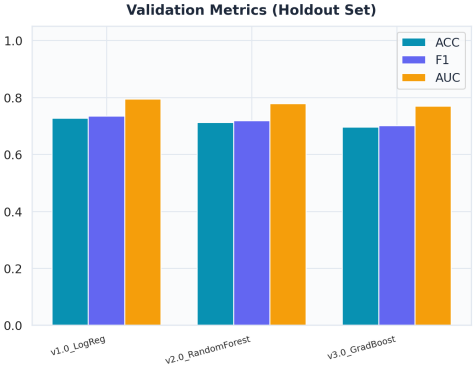
Days 0–30: Stable baseline

Days 30–60: Gradual drift onset

Days 60–90: Severe distribution shift

Model Registry & Validation

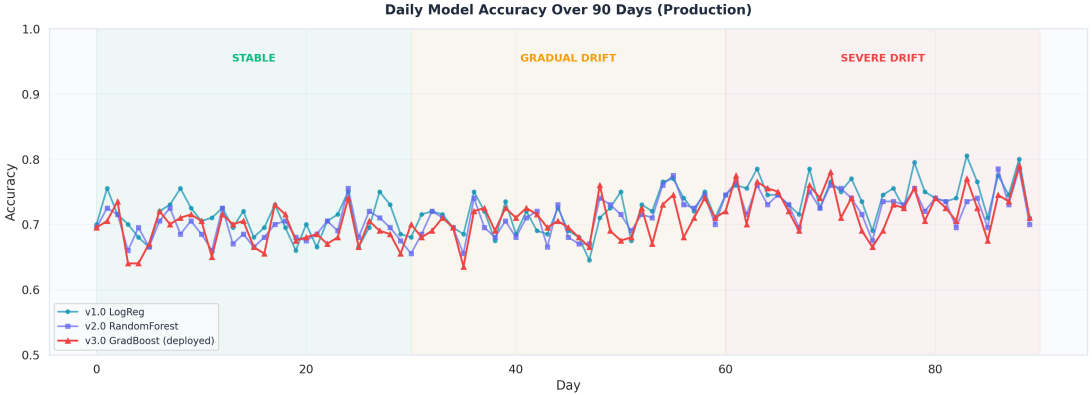




Three Production Versions

Version	Algorithm	Hash
v1.0	Logistic Regression	f34ee29a

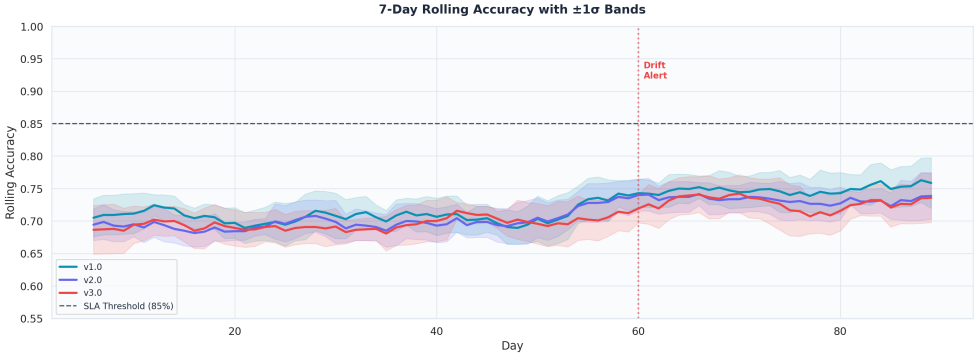
Daily Accuracy Tracking



Drift Detected

All three model versions degrade after Day 60. The deployed v3.0 (Gradient Boosting) drops from ∼92% to ∼73% accuracy — below the 85% SLA threshold.

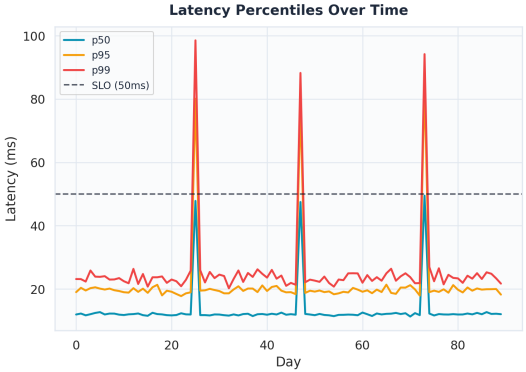
Rolling Performance & SLA Compliance

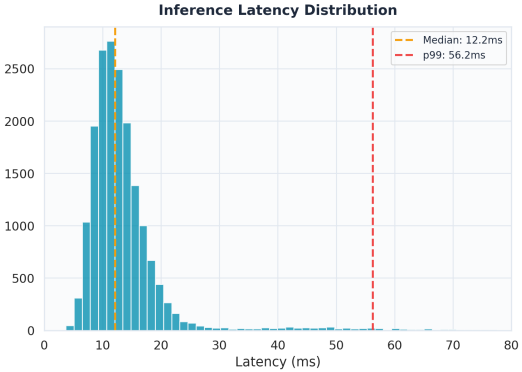


7-Day Rolling Metrics

Smoothed view reveals the structural break at Day 60. SLA breach triggers an automated alert and initiates the retraining workflow.

Inference Latency Monitoring

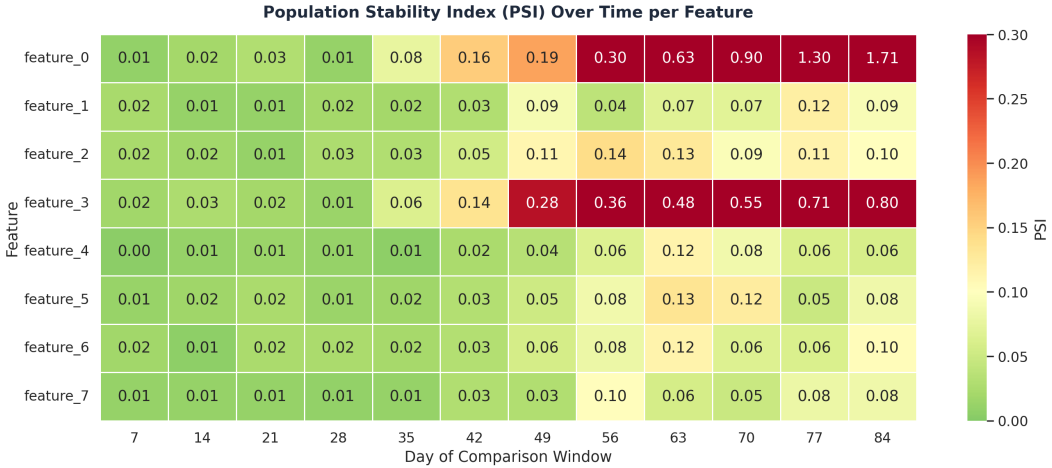




Latency SLOs

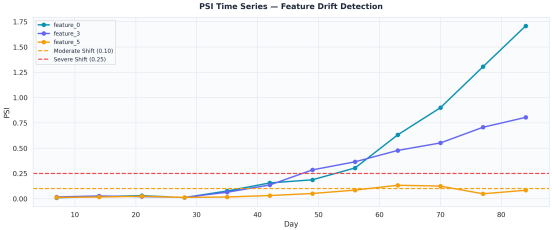
Avg latency	13.8 ms	p99	56.2 ms
SLO target	50 ms	Incidents detected	3

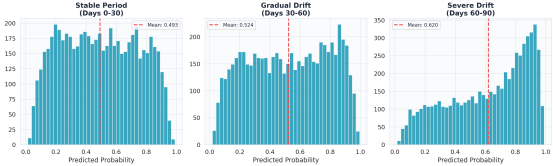
Population Stability Index (PSI)



PSI Thresholds

PSI Time Series & Prediction Drift

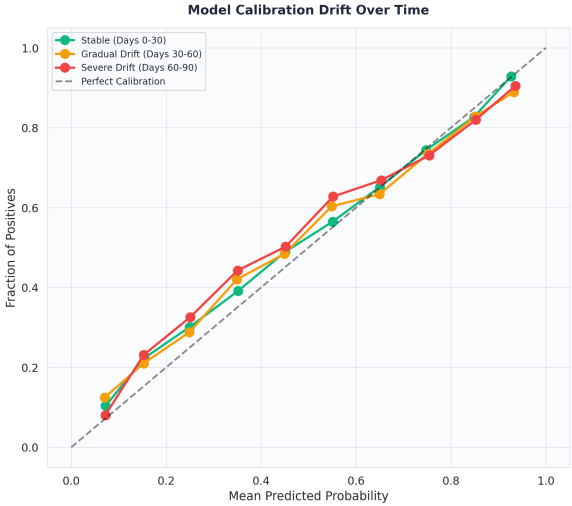




Drift Signals

Feature 0 and Feature 5 exceed the critical PSI threshold by Day 65. Prediction distribution shifts toward extremes during the severe drift period — a hallmark of model overconfidence.

Calibration Drift

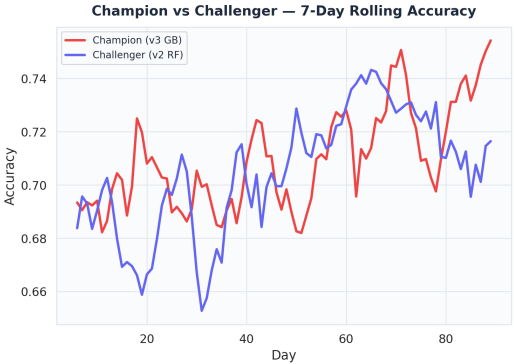


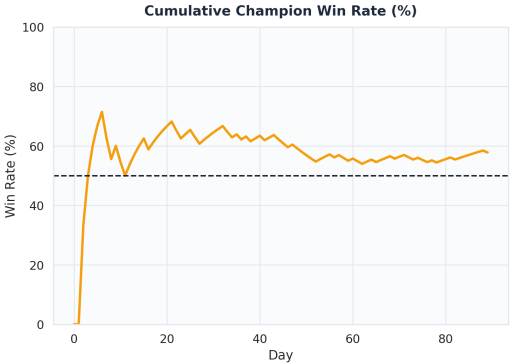
What Calibration Drift Means

A well-calibrated model's predicted probabilities match observed frequencies. As drift progresses:

- Stable period: near-perfect calibration
- Gradual drift: minor deviations
- Severe drift: model becomes **systematically overconfident**

Champion vs Challenger in Production

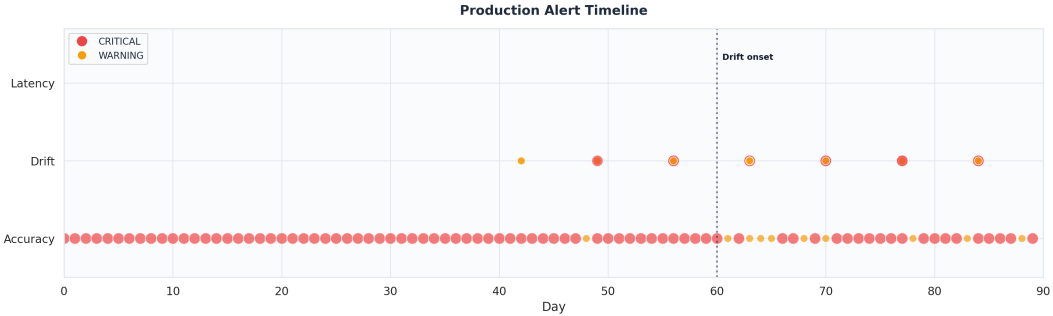




Live A/B Testing

50/50 traffic split between v3 (champion) and v2 (challenger). Cumulative win rate informs the deployment decision: if challenger consistently wins, promote it.

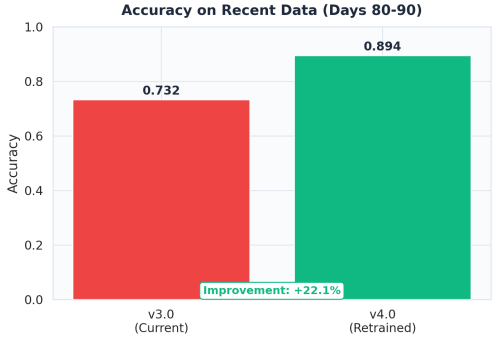
Production Alert Timeline

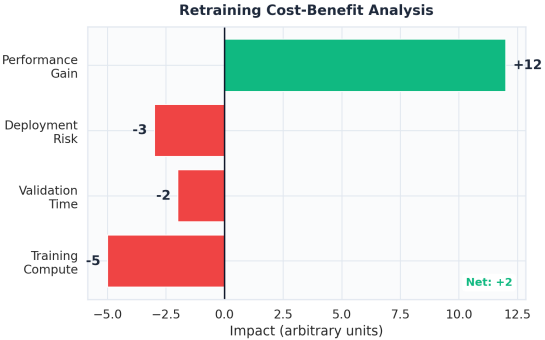


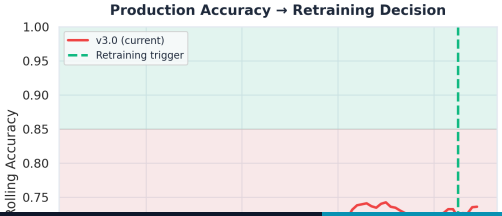
116 Total Alerts Triggered

Multi-signal alerting from accuracy degradation (SLA), drift detection (PSI), and latency spikes. Critical alerts cluster after Day 60, confirming the structural shift.

Retraining Decision & Rollout





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RETRAINING RECOMMENDATION

Status:  ⚠  RETRAIN RECOMMENDED

Triggers Activated:
  ✓ Accuracy below SLA (85%)
  ✓ PSI > 0.25 on multiple features
  ✓ Calibration drift detected
  ✓ A/B test confirms degradation

Proposed Action:
  → Deploy model v4.0 (Gradient Boosting)
  → Trained on recent 20-day window
```

Summary & Production Readiness

Capabilities Demonstrated

- Multi-version model registry
- Real-time monitoring (acc, latency)
- PSI-based drift detection
- Calibration tracking
- A/B testing framework
- Cost-benefit retraining triggers
- Canary rollout strategy

Production Stack (Future)

1. **Serving:** FastAPI + Docker + Kubernetes
2. **Tracking:** MLflow / W&B
3. **Feature store:** Feast / Tecton
4. **Observability:** Prometheus + Grafana
5. **Data quality:** Great Expectations
6. **CI/CD:** GitHub Actions

Thank You

Questions & Discussion

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